

Proof. Theorem (Jensen's Inequality). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex. For real numbers $x_1, \dots, x_n$ and weights $\lambda_1, \dots, \lambda_n$ satisfying

$$\lambda_i \geq 0 \ (i = 1, \dots, n), \quad \sum_{i=1}^n \lambda_i = 1,$$

we have

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (1)$$

1. Base case $n = 2$. Write $\lambda_1 = t$ and $\lambda_2 = 1 - t$ with $t \in [0, 1]$. Convexity of f gives

$$f(tx_1 + (1 - t)x_2) \leq t f(x_1) + (1 - t) f(x_2). \quad (2)$$

Taking $t = \lambda_1$ yields (??) for $n = 2$.

2. Induction hypothesis. Assume that (??) holds for every collection of k points with any non-negative weights summing to 1, for all $k \leq n - 1$ (with $n \geq 3$).

3. Induction step (the case of n points). Let $\lambda_1, \dots, \lambda_n$ be as in the statement and set

$$\mu := \sum_{i=1}^{n-1} \lambda_i.$$

3.1 Degenerate sub-case $\mu = 0$. If $\mu = 0$ then $\lambda_i = 0$ for $i = 1, \dots, n - 1$ and $\lambda_n = 1$, so both sides of (??) reduce to $f(x_n)$ and the inequality holds with equality.

3.2 Non-degenerate sub-case $\mu > 0$. Define normalised weights for the first $n - 1$ points

$$\alpha_i := \frac{\lambda_i}{\mu}, \quad i = 1, \dots, n - 1,$$

so that $\alpha_i \geq 0$ and $\sum_{i=1}^{n-1} \alpha_i = 1$. Set

$$y := \sum_{i=1}^{n-1} \alpha_i x_i.$$

Then $\mu y = \sum_{i=1}^{n-1} \lambda_i x_i$.

Applying the induction hypothesis to the points $(x_i)_{i=1}^{n-1}$ with weights (α_i) yields

$$f(y) \leq \sum_{i=1}^{n-1} \alpha_i f(x_i). \quad (3)$$

Multiplying (??) by the positive scalar μ gives

$$\mu f(y) \leq \sum_{i=1}^{n-1} \lambda_i f(x_i). \quad (4)$$

Since $\mu + \lambda_n = 1$ and $\mu \in [0, 1]$, convexity applied to the two points y and x_n with weights μ and λ_n yields

$$f(\mu y + \lambda_n x_n) \leq \mu f(y) + \lambda_n f(x_n). \quad (5)$$

Because $\mu y = \sum_{i=1}^{n-1} \lambda_i x_i$, the left-hand side of (??) is $f(\sum_{i=1}^n \lambda_i x_i)$. Substituting (??) into the right-hand side of (??) we obtain

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^{n-1} \lambda_i f(x_i) + \lambda_n f(x_n) = \sum_{i=1}^n \lambda_i f(x_i),$$

which is exactly (??) for n points. This completes the induction.

4. Equality condition. Equality in (??) can occur only when equality holds in every convexity application used above.

- In the base case $n = 2$, equality in (??) holds precisely when either $t = 0$, $t = 1$ (i.e. one weight is zero) *or* f is affine on the segment $[x_1, x_2]$ (the latter includes $x_1 = x_2$ when f is strictly convex).
- In the induction step, the second convexity application (??) is an equality iff either $\mu = 0$ or $\lambda_n = 0$ (so the combination reduces to a single point) *or* f is affine on the segment $[y, x_n]$.

Consequently, equality in (??) holds exactly in one of the following situations:

1. **Trivial concentration of weight:** there exists k with $\lambda_k = 1$ and $\lambda_i = 0$ for $i \neq k$; both sides equal $f(x_k)$.
2. **All positively-weighted points coincide:** if $\lambda_i > 0$ and $\lambda_j > 0$ then $x_i = x_j$. The convex combination reduces to that common value, giving equality.
3. **Affineness of f on the relevant convex hull:** the restriction of f to the convex hull of $\{x_i : \lambda_i > 0\}$ is affine (linear). This includes the case where f is globally affine.

These possibilities are exhaustive; each indeed yields equality in (??).

Thus for any convex f , any real numbers $x_1, \dots, x_n$, and any non-negative weights $\lambda_1, \dots, \lambda_n$ summing to 1, Jensen's inequality (??) holds, with equality precisely in the cases described above. $\square$ $\square$